

# Geralt Heart

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## SUMMARY

Graduate student at The University of Colorado Boulder pursuing Computer Science with a focus on Machine Learning, Data mining, Network Systems, and Computer Vision. Contributor to AI training and LLMs through Reinforcement learning. Experienced in Python and SQL with a passion for web development and deployment.

## WORK EXPERIENCE

### Software Engineering Intern

*May 2026 – Aug 2026*

Athena Manufacturing LP

Austin, TX

- Designed and deployed an internal Kanban-style automated scheduler parsing data from Microsoft SQL Server into a Python application utilizing pyodbc, PyQt6, Pandas, and JSON
- Created a local database for data persistence and enforce authentication database access for networking between different users. Utilized pymssql to validate user identity when accessing M2M data.

### AI Trainer

*Feb 2026 – Present*

Handshake

Austin, TX

- Evaluated AI-generated content for factual accuracy and monitored hallucinations. Supported Reinforcement Learning from Human Feedback pipelines to optimize performance across projects.

## PROJECTS

### Regression Testing Framework | *python, pytest, GitHub Actions*

[github.com/GHeart01/tile-gps-regression-testing](https://github.com/GHeart01/tile-gps-regression-testing)

- Designed 52 automated unit and regression tests covering 4 system modules (TILE, EMQuest GPS, Spectra Python, and Aurora systems) achieving 100% test validation with execution time under 0.1 seconds.
- Validated Android GPS module functionality including location accuracy, signal strength, and RTK/DGPS compatibility using integration tests for data persistence, serialization, JSON validation, and configuration.
- Created CI/CD pipeline using GitHub Actions with automated testing, coverage analysis, & multi-version support.

### Pricing Web Scraper | *PostgreSQL, Python, BeautifulSoup, REST API*

[github.com/GHeart01/Web-Scraper-Clothing](https://github.com/GHeart01/Web-Scraper-Clothing)

- Implemented a PostgreSQL relational database schema for indexing the price of men's clothing across 4 different major retailers.
- Utilized rate limiting to manage anti-scraping protections (403 errors) and evaluated performance against REST API alternatives. Created master scraper controller that aggregates results from multiple sources, calculates price comparisons, and exports normalized JSON data compatible with PostgreSQL

### Customer Segmentation & Clustering Analysis | *NumPy, Pandas, seaborn, matplotlib, TensorFlow*

[github.com/GHeart01/Wholesale\\_Customer\\_Clustering](https://github.com/GHeart01/Wholesale_Customer_Clustering)

- Built a customer segmentation pipeline using K-Means and Hierarchical Clustering to identify purchasing behavior patterns. Applied hyperparameter tuning across several different models to evaluate performance.
- Evaluated clustering quality using silhouette scores between multiple cluster counts, provided heatmap/pair plot visualizations to demonstrate feature relationships.

## EDUCATION

### University of Colorado Boulder | GPA: 3.9/4.0

*Expected May 2027*

Master of Science in Computer Science

Boulder, CO

### University of Texas Rio Grande Valley | GPA: 3.5/4.0

*July 2017 – May 2020*

Bachelor of Science in Applied Mathematics

Edinburg, TX

## SKILLS

**Programming Languages:** Python, SQL, C++, JavaScript, MATLAB

**Frameworks & Libraries:** React, Node.js, pytest, Three.js, Pandas, NumPy, scikit-learn, TensorFlow, Keras, SciPy

**Databases:** PostgreSQL, MySQL, MS SQL Server

**Cloud & DevOps:** AWS Lightsail, Docker, Kubernetes